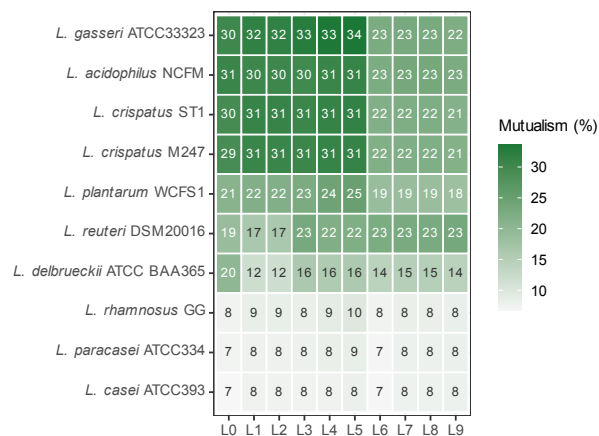


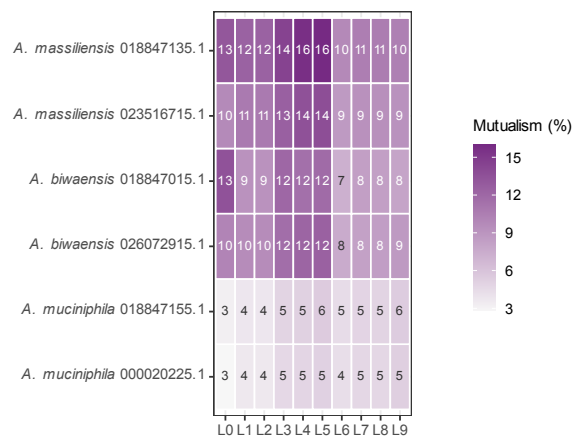
A

Lactobacillus × gut, strain-resolved



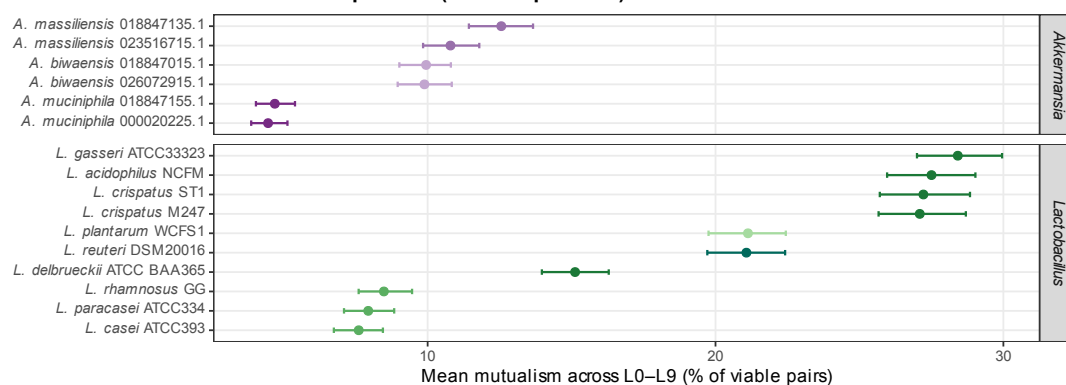
B

Akkermansia × gut, strain-resolved



C

Per-strain cooperation (bootstrap 95% CI)

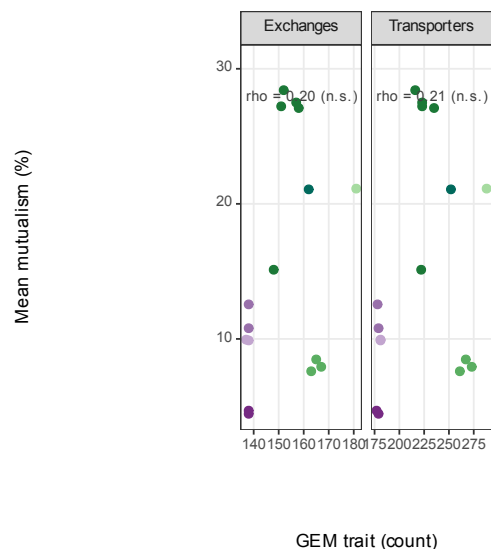


● *A. biwaensis* ● *A. massiliensis* ● *A. muciniphila* ● *Lactocaseibacillus* ● *Lactiplantibacillus* ● *Lactobacillus* ● *Limosilactobacillus*

D

Capacity does not predict cooperation

highest-capacity *Lactocaseibacillus* cooperate
least — no monotonic relationship



E

Cooperation by partner phylum

